University of Glasgow

English Communication & Engineering Career Skills

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Lecture 3: ECECS Assessment & Grand Challenges

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Aims

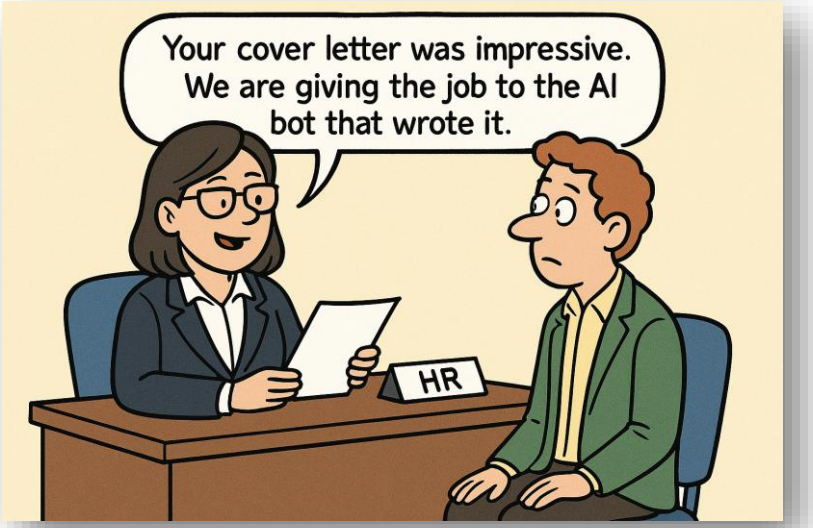
- Learn about the course assessments
- Learn about the NAE Grand Challenges and how they link to the assessments
- Consider the relationship between Engineering Grand Challenges and local issues

A quick recap: cover letters!

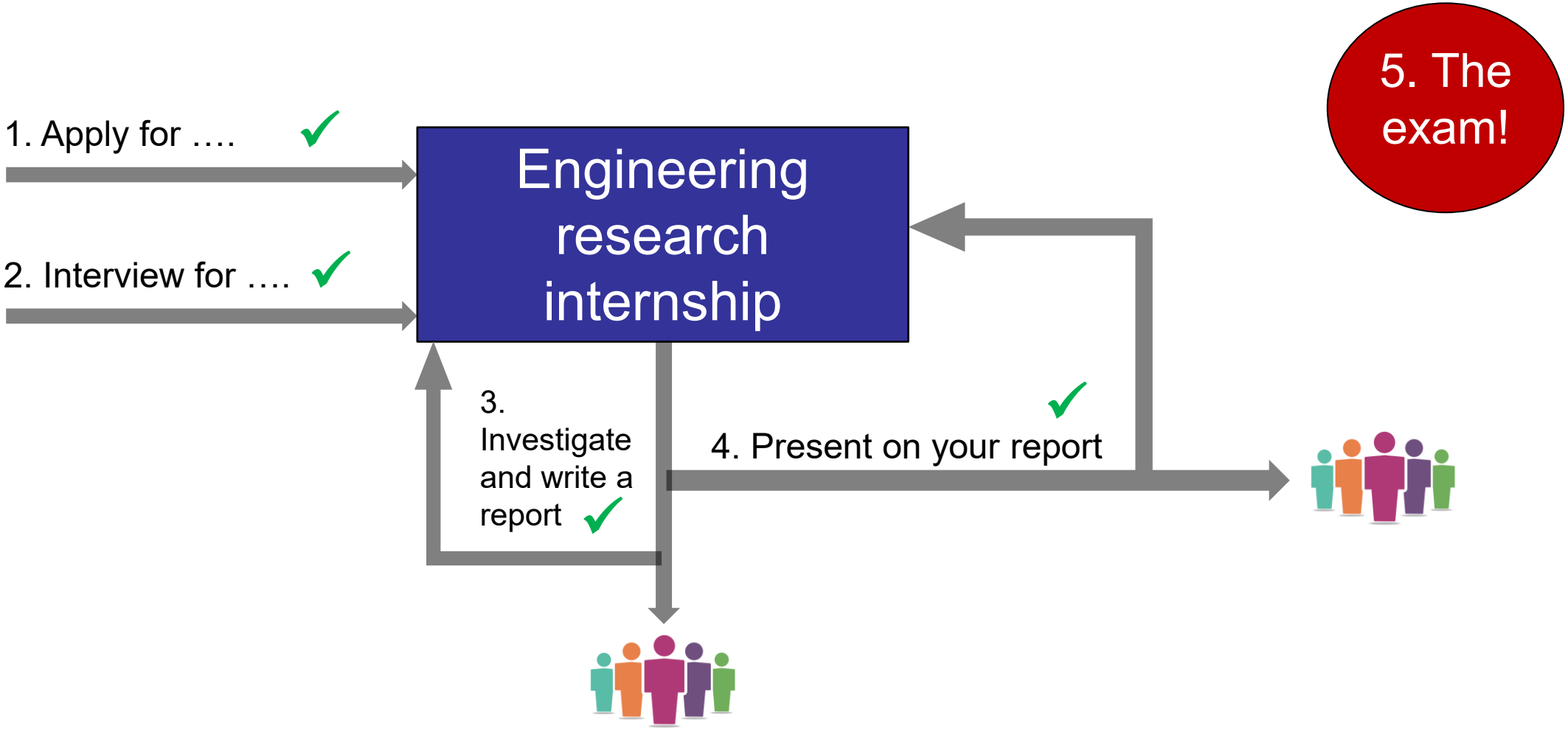
Don’t

- exaggerate or overstate your experience
- beg for the job
- use AI to generate your letter! Because AI cover letters ….
    - use generic, clichéd phrases, e.g. “I’m excited to apply for this opportunity at **your esteemed company** … I have **a proven track record** in … **Beyond academics** …”
    - are full of **buzzwords**, e.g. “I have a **results-driven mindset**”
    - don’t mention the company or team
    - don’t demonstrate *your* personality, *your* motivation, *your* effort

HR managers will often reject candidates because they recognise the signs of AI generation



(Almost) all roads to ECECS success relate to an internship position



Assessments 1 & 2: Portfolio and Group Interview

Assessment	Weighting
Portfolio **Set up in Wk 2 (CS 3)**	10%
Group interview (part of oral assessment) **Set up in Wk 3 (CS 5)**	10%

simulation of real world tasks

Portfolio

You will be asked to apply for an **internship**:

- CV (includes a **career objective** & **project proposal**)
- Cover letter: what's your motivation?

Group interviews

- Based on the same internship task
- Interviewee role: respond to questions appropriately, show interactive skills

Assessment 3: Written Assignment

You are currently engaged **on an internship** at the company **XXXX**. As part of your internship training and evaluation, you need to investigate an **Engineering Grand Challenge** in the context of a **specific locality** and produce a **report**.

- You are given the Challenge and 2 sources: you choose the locality/issue
- You research the issue, identify 2 solutions and evaluate them
- The audience includes non-specialists

simulation of
real world tasks

Assessment	Weighting
Written assignment	20%
Set up in Lectures Wk 4	

Assessment 4: Group Presentation

You have been asked by your supervisor to deliver a presentation with one other intern on the theme of your internship research project for a networking event that the company is running.

The audience will be non-specialists from other organisations. Your presentation is for a maximum of 10 minutes (with an additional three minutes for questions and answers) and must be of interest and engaging to a general audience.

simulation of
real world tasks

Assessment	Weighting
Presentation **Set up in Lectures Wk 9**	20%

Assessment 5: Final Exam

You will complete tasks based on two texts.

You will cover these skills:

- Writing an abstract for a research article
- Answering reading questions on a general engineering-related article
- Writing a summary and an evaluation based on the general article

simulation of
real world tasks

Assessment	Weighting
Final Exam **Set up in Week 13**	40%

Context: The NAE Grand Challenges

grand (adj)

'important and large in degree'

She has all kinds of grand ideas

https://dictionary.cambridge.org/dictionary/english/grand

'sweeping in ambition or conception'

a grand scheme to build a canal across the desert

https://www.thefreedictionary.com/grand

1. impressive in size, appearance, or general effect: *grand mountain scenery.*
2. stately, majestic, or dignified: *In front of an audience her manner is grand and regal.*
3. highly ambitious or idealistic: *grand ideas for bettering the political situation.*
4. magnificent or splendid: *a grand palace.*

https://www.dictionary.com/browse/grand

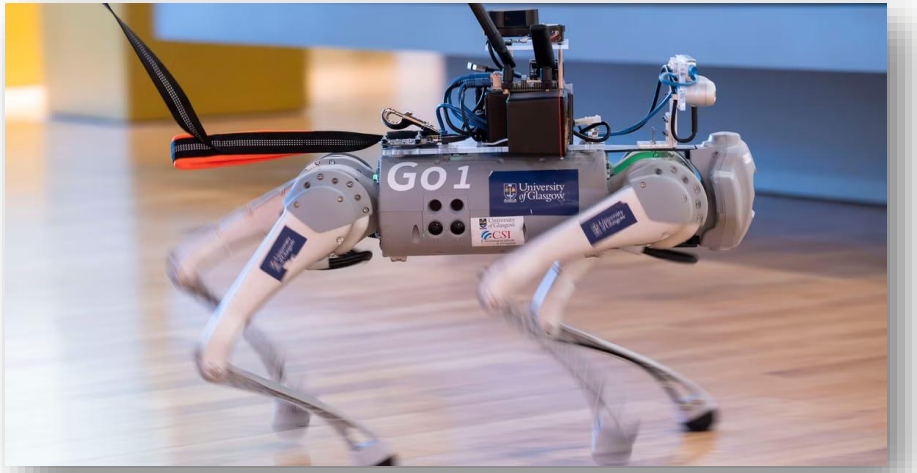
So a grand challenge is …

- a fundamental problem or issue in engineering
- complex
- a global issue – but with specific **local** aspects
- of importance not just to engineers, but also to society in general
- relevant to engineers from all subdisciplines

Solutions will impact positively globally – but will need to be **tailored** to localities

FOR EXAMPLE?

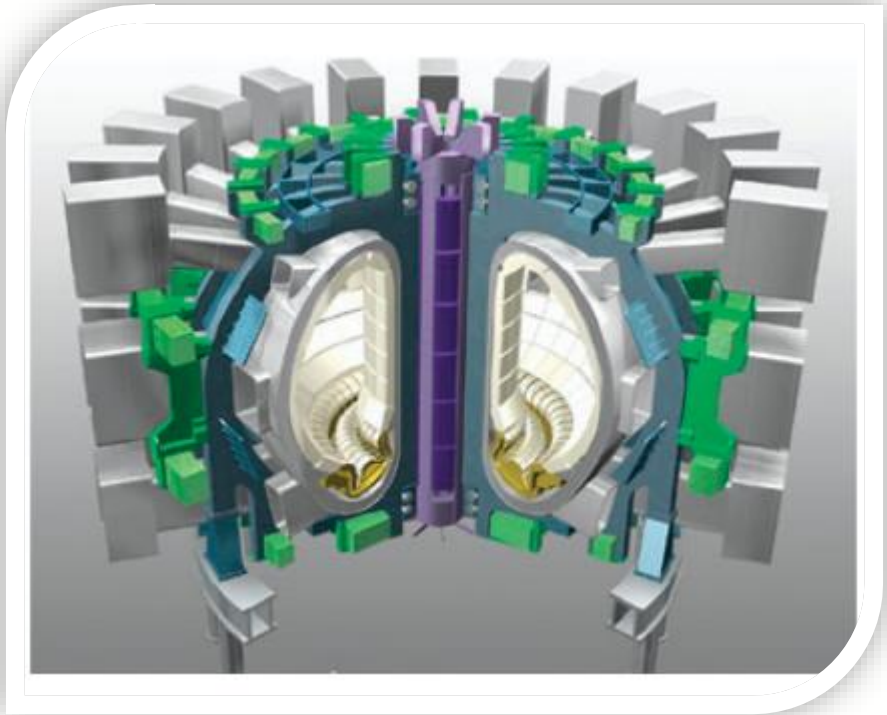
Which grand challenge could this invention relate to?



University of Glasgow - University news - Paw-sitive reception for robot guide dog could lead to new assistive technology

The NAE Grand Challenges are …

1. Make solar energy economical



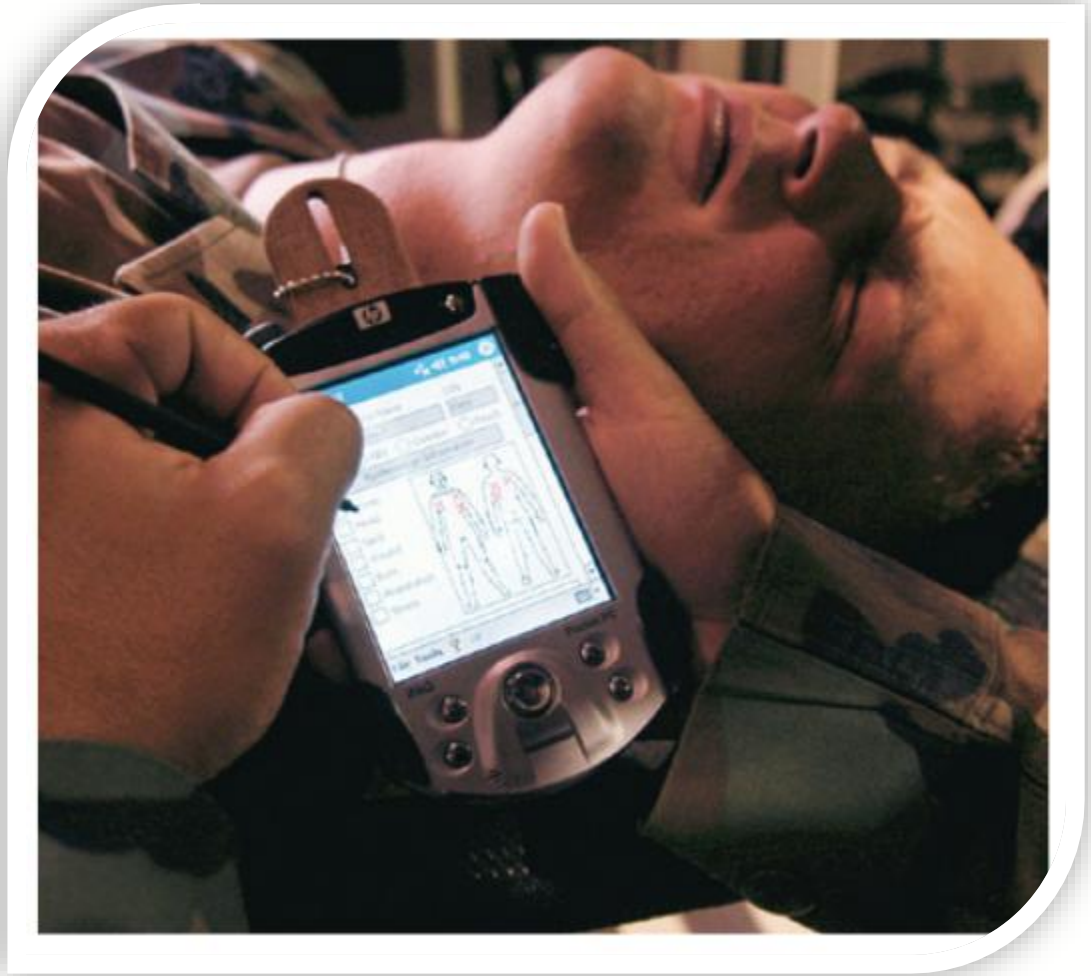
2. Provide energy from fusion

3. Develop carbon sequestration methods

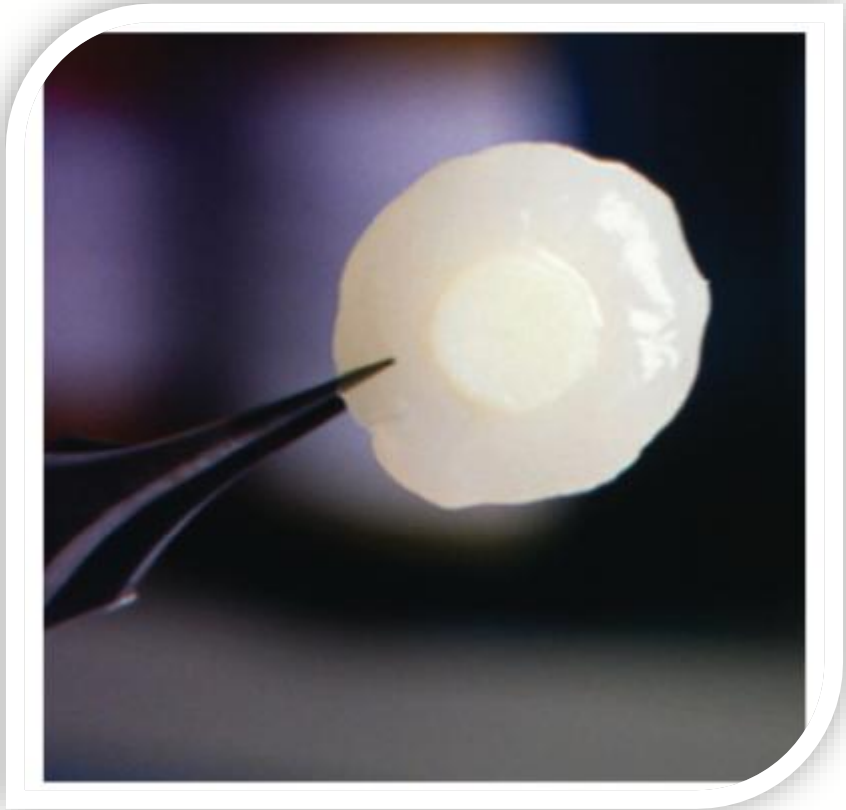
4. Manage the nitrogen cycle

5. Provide access to clean water

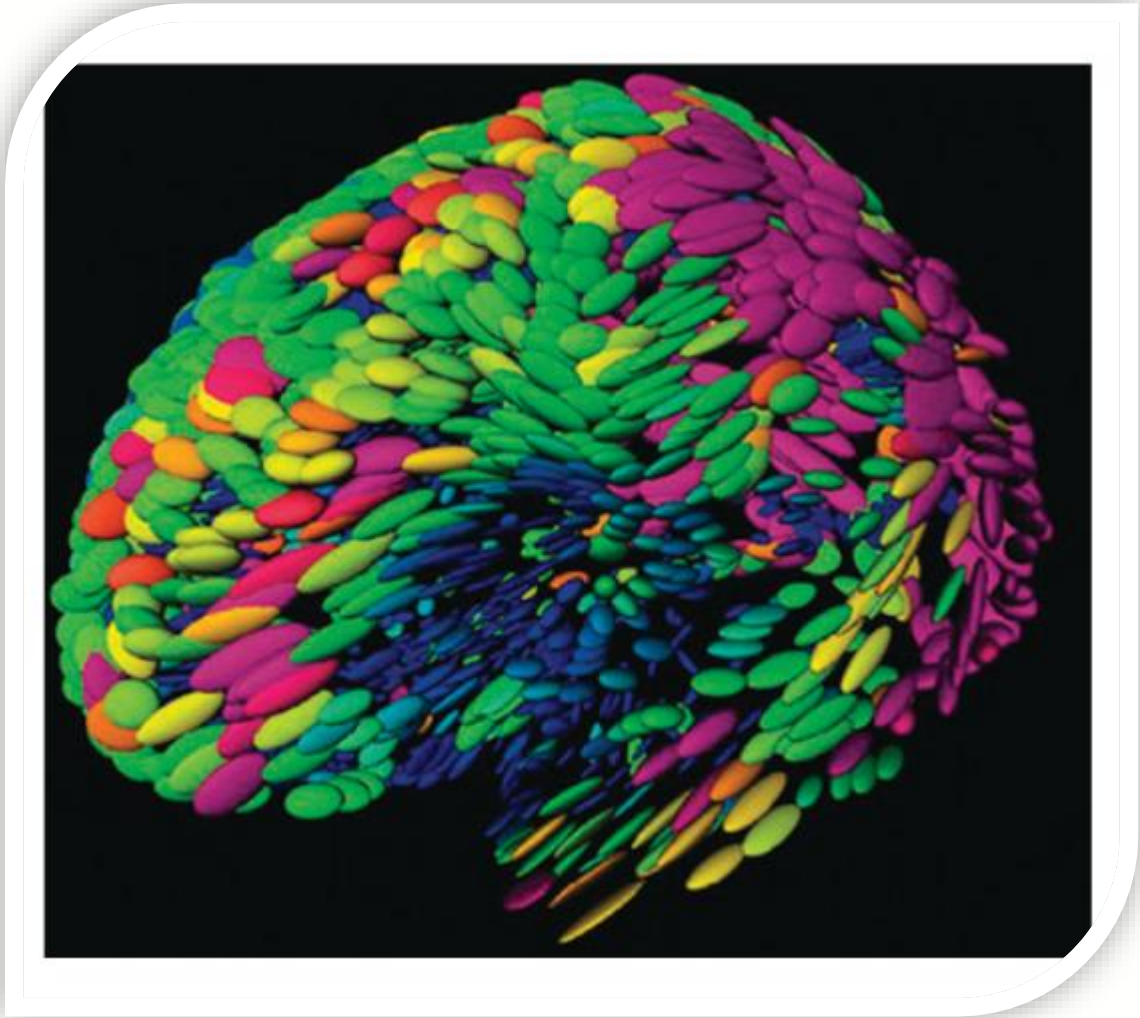
6. Restore and improve urban infrastructure



7. Advance health informatics



8. Engineer better medicines



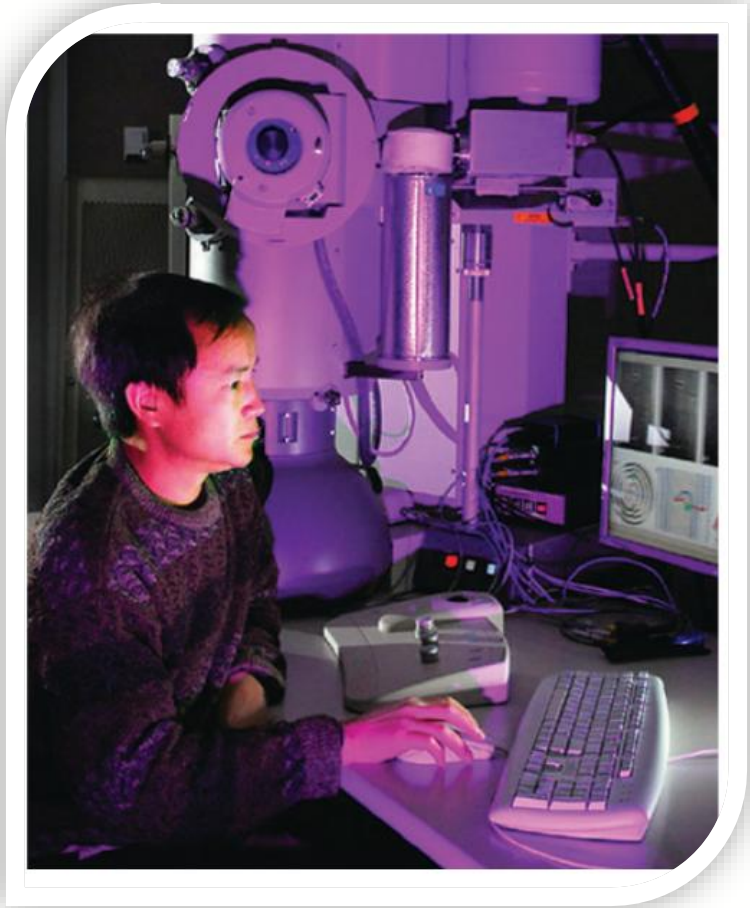
9. Reverse-engineer the brain

10. Prevent nuclear terror

11. Secure cyberspace

12. Enhance virtual reality

13. Advance personalised learning



14. Engineer the tools of scientific discovery

Which NAE Grand Challenge is most interesting to you?

Make solar energy economical	Provide energy from fusion	Develop carbon sequestration methods	Restore and improve urban infrastructure	Secure cyberspace	Advance health informatics	Engineer the tools of scientific discovery	Enhance virtual reality	Reverse-engineer the brain	Advance personalized learning
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Waiting for participants

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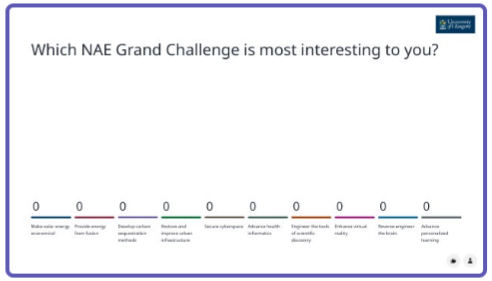
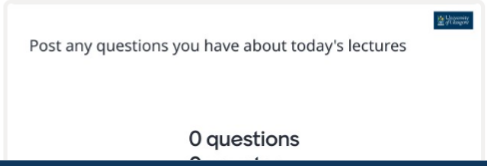
Poll: Most Interesting NA...

Choose a slide to present





Issues for the Grand Challenges

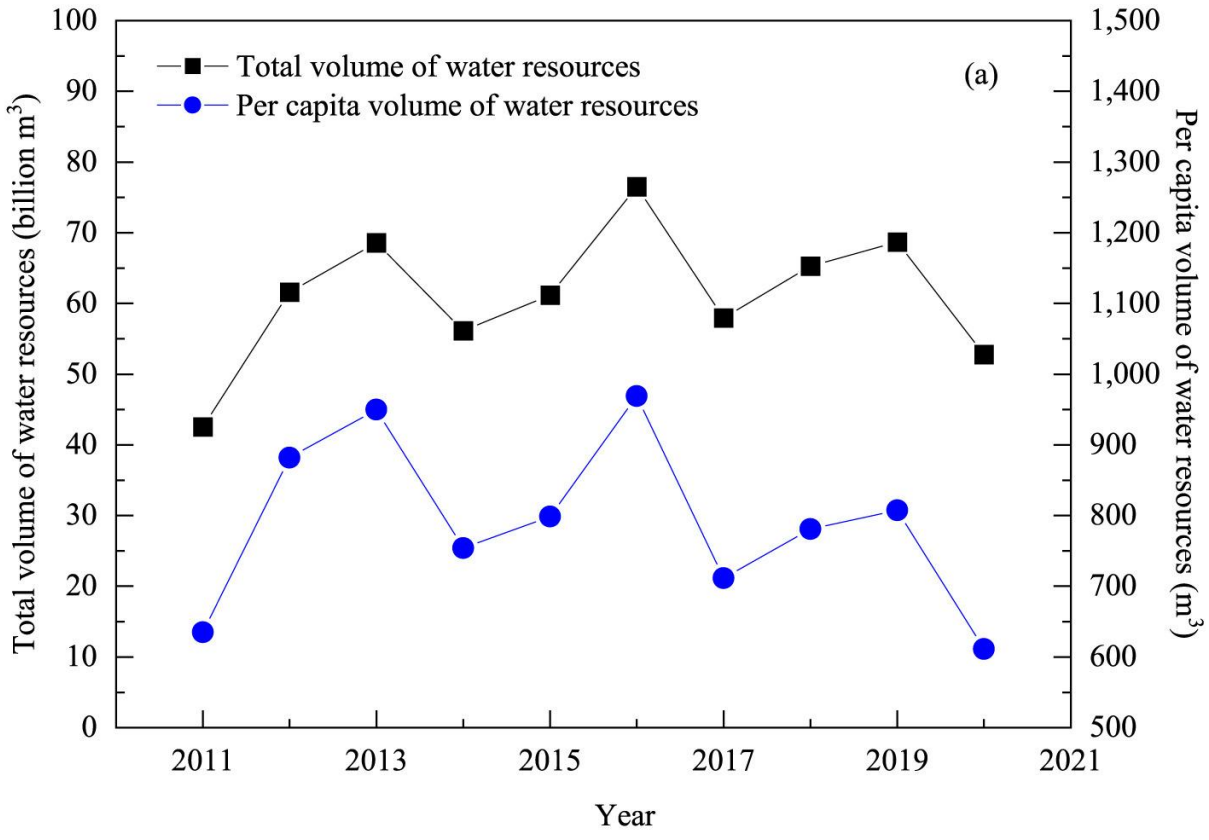
- Do these challenges encompass everything that's important for engineers and society? Where would the robot dog fit into this picture?
- Are some of these Challenges out of date?
- Are some of these more important to the USA than to other nations?
- What are the Grand Challenges from a Chinese perspective?

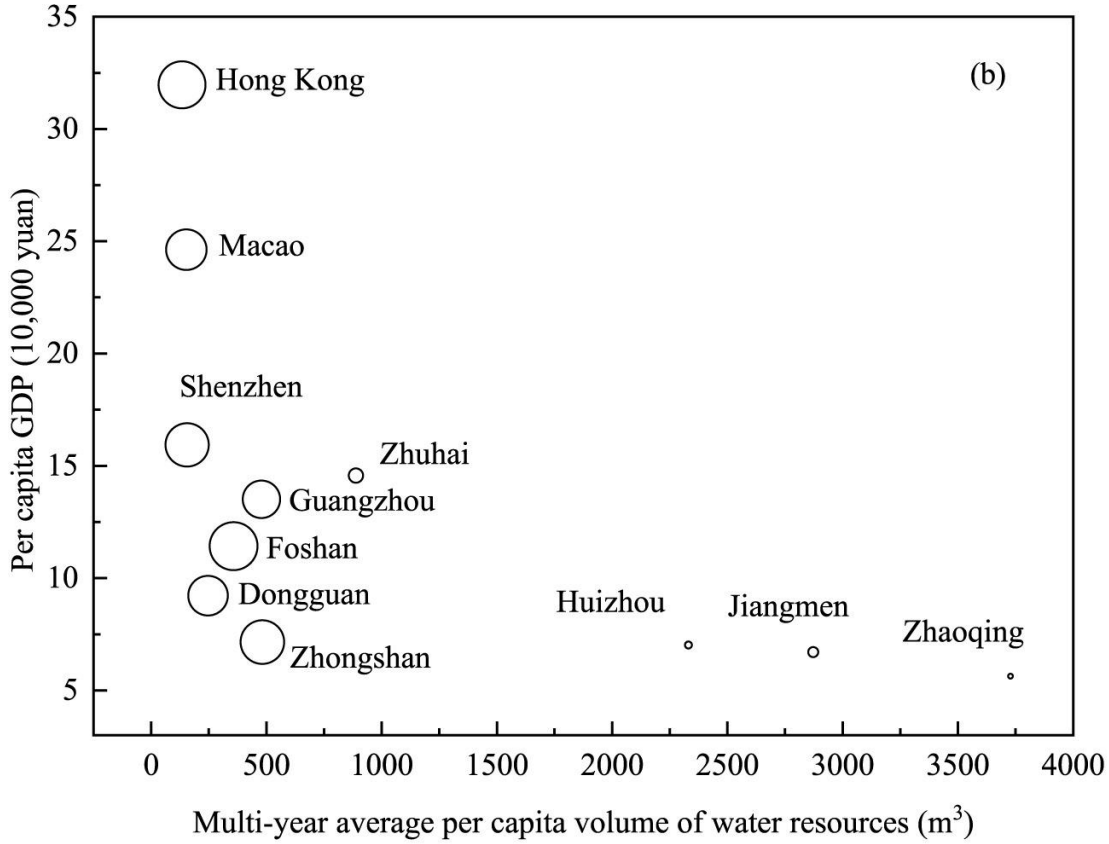
Top 10 Engineering Challenges: China

1. Integrated Design-Simulation-Manufacturing Algorithms
2. Scalable Deep-Sea Mining and Eco-Friendly Technologies
3. Coordinated Water Resource Management
4. Smart and Simplified Network Architecture for 6G
5. Biomanufacturing of Complex Organs
6. Integrated Mining of Coal and Associated Resources
7. Next-Generation Low-Cost Carbon Capture and Storage
8. Aircraft Energy and Thermal Management Systems
9. Biomanufacturing of Food Ingredients
10. Data-Driven Drug Discovery Using Clinical Multi-Omics

https://voc-gj.cast.org.cn/index/info?api=GwArticle&id=40360

Coordinated Water Resource Management: The Greater Bay Area of China





Wang, Y., Huang, C., Wu, G., & Wang, W. (2022). Status and challenges of water resources and supply in the Guangdong-Hong Kong-Macao Greater Bay Area (GBA) of China. *Water Cycle*, *3*, 65–70. https://doi.org/10.1016/j.watcyc.2022.05.001

Out-of-class work

Homework:

- Review your notes and the PPT slides.
- Read further about the NAE Grand Challenges for Engineering here: https://www.nae.edu/170270/Reports#tabs

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MINOR HEAT

Aims

- Gain insights into the Grand Challenge “Make Solar Energy Economical”
- Develop listening skills in relation to a lecture on this topic
- Learn about local solutions to making solar energy economical

NAE GC1: Make solar energy economical

- In 2008, the cost of solar energy in the US was around 18-30c per kWh
- Commercial solar cells had an efficiency of 10-20%
- Today, the cost has fallen as low as 4c per kWh and efficiency can reach 25%
- Alternatives to solar cells can reach 30-40% thermal-electric conversion, with better storage possibilities
- So has this challenge been solved, or do issues with solar remain?

Key vocabulary and acronyms (1)

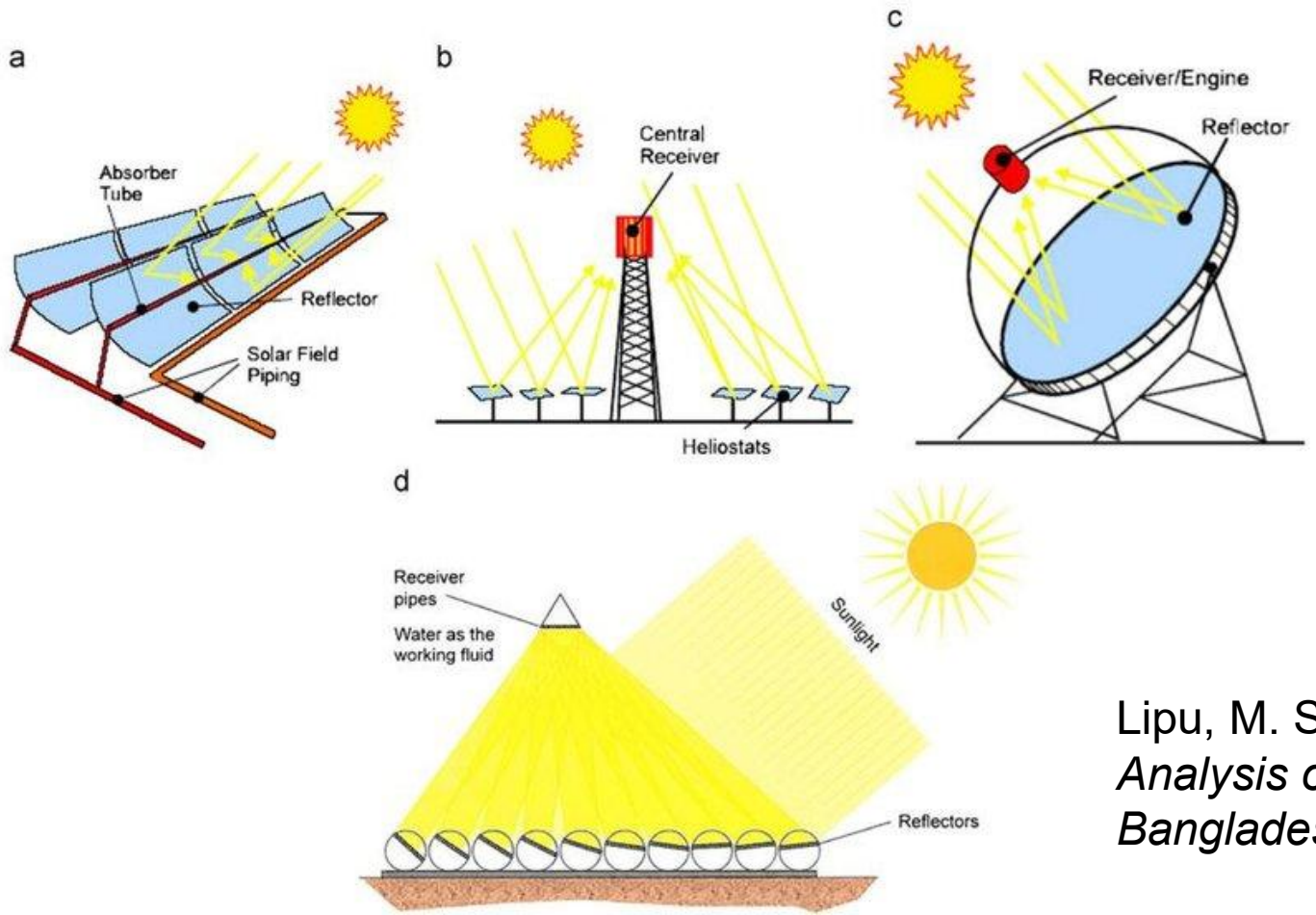
- The **DOE**: The US Department of Energy
- **Sunshot**: a DOE initiative to reduce the cost of solar (to 3c per kWh for 2030)
- **Tarriffs (n)** – taxes on foreign imports. US president Trump imposed high tariffs on imported solar cells in 2018
- **Deployment** (n) / **to deploy** (v) – “the action of bringing resources into effective action” (Oxford) – also **to dispatch**
- **Utility** (n) = use/value; **utility scale** = large scale commercial energy (more efficient, higher value)
- **To dial back** (v) = to reduce

Key vocabulary and acronyms (2)

Do you know or can you guess the meaning of these solar-related acronyms?

- **PV** — Photovoltaic cells
- **CSP** — Concentrated Solar Power
- **O&M costs** — Operation & Maintenance

CSP technologies



a. Parabolic <u>trough</u>
b. Power tower
c. Parabolic dish system
d. Linear Fresnel reflector connectors

Lipu, M. S. H., & Jamal, T. (2013). *Techno-economic Analysis of Solar Concentrating Power (CSP) in Bangladesh*. *2*(5).

Lecture: Low cost solar energy is here, what's next? Dr. Nathan Siegel, Bucknell University

Listen and take notes by hand.

Try to identify the **main sections** of the lecture and make headings

Add detail under the headings: what are Dr. Siegel's main points?

There are subtitles to help you!

Feedback – my notes

Introduction

Solar has come down in cost
2 issues in making it econom.: ↓ expensive, ↑ useful (e.g. make available at night)
Too $$ in 2008; now (2019) 4c per kWh
Deployment up (despite tariffs) – lot of growth in CSP (e.g. power tower)
Now more PTs (less troughs) with 12 hours storage – e.g. Crescent Dunes in Nevada

Costs - PV

Location matters – Daggett (California – desert)
Module cost can reduce
Increase lifetime
Reduce O&M
All these help meet Sunshot target (3c per kWh) – but Siegel thinks better is poss.

Costs – CSP

Sunshot target is higher (5c per kWh) because storage is included and that brings more value
Key is reducing heliostat cost to <$50 per sq m.

Efficiency

… brings flexibility – more efficient systems = more to spend on module costs
Key is to >> efficiency but keep module cost low
In PV – multi-junction cells can help
Layered semiconductors with diff bandgaps – but in 2019 this is too exp.
In CSP – you need ↑temps
BUT losses increase! 700° is close to "sweet spot"
Cutting collection losses also important – getting light to receiver is lossy – opp. for innovation – e.g. use a parabolic dish

Utility/value

Make it more **dispatchable** = storage but more than that
The duck - ↑ solar production in daylight, other generation has to dial back; but some (e.g. nuclear) can't be turned off; gas turbines can; but if u turn off solar, u lose value
↓ Solar at night, but demand ↑↑ = problems for the grid
∴ Storage is essential to deploy at night (also grid flexibility)
Batt. – not the answer bc life cycle & when sunlight ↓ (e.g. winter – seasonal stor.) – cost of batt. not worth it

What happens next?

Question: imagine we have 1c per kWh but only for 6 hrs – what are you going to do?
Poss: making fuel from sun; desal.; charging EVs in day; but systems not opt. for intermittent operation
Systems need to adapt to this reality

Notes on lecture style

This lecture style is typical of a US academic – technical, authoritative **but** relaxed, informal delivery!

- Humour: “I almost missed the chair there!” … “Hope I hit the right button … oh man … There we go!”
- Most frequent linking expressions: **and**, **but**, **so -** little or no use of formal written linkers like **moreover**, **however**, etc.
- Shortened sentences to help understanding: “So you saw those mirrors. Mirrors are made of glass and steel. Glass and steel cost a certain amount of money.”
- Informal expressions & phrasal verbs: “That would be a **big win**” … “Take this with **a grain of salt**” (= don’t believe this too easily) … “You can **figure out** …” … “**dial back** …” … “that’s **pretty** important” … “this results in a **sort of** maximum” … “That’s **not fun** for the grid”.

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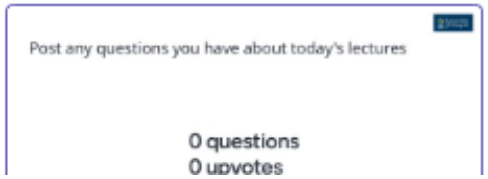
Poll: Most Interesting NA...

Choose a slide to present



Post any questions you have about today's lectures

No questions from the audience!

Incoming questions will show up here so that you can answer them one by one.

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Waiting for participants

Out-of-class work

Homework:

- Review the work you have done so far.
- Watch the whole lecture and interview about solar on Moodle to develop your extended listening skills

Specifically:

- Compare your lecture notes with mine. How can you improve these? Rewatch the lecture and try to add detail.
- Complete the Moodle task I will send tonight: watch the final part of the video (interview).
    - Which examples are given of **local** solutions to issues of solar generation and storage? What makes these solutions **local**?
    - Can you give me any examples of local solutions to solar generation and storage from China?